

SUMMARY

Data Science and Engineering undergraduate specializing in full-stack development, scalable backend architecture, and secure system design for high-traffic platforms. Strong analytical problem solver recognized through competitive programming achievements (Codeforces Specialist, 700+ LeetCode solutions), highly dedicated to building efficient, user-centric software solutions that drive measurable business impact.

EDUCATION

• National Institute of Technology (NIT), Jalandhar	2023 – Present
• <i>B.Tech. in Data Science and Engineering Coursework: ML, NLP, Systems Design, Cloud</i>	<i>CGPA: 8.29</i>
• Gobindgarh Public School, Mohali	2021 – 2023
• <i>Class 11–12 (Science Stream)</i>	<i>89%</i>

SKILLS

- **Programming Languages:** C, C++, JavaScript, TypeScript, Python, Golang
- **Web & Full-Stack:** React.js, Next.js, Node.js, Express.js, Tailwind CSS, WebSockets, tRPC, gRPC, Prisma ORM
- **Cloud, Databases & Tools:** PostgreSQL, MongoDB, MySQL, Docker, Kubernetes, AWS, Linux, Git, REST APIs, Microservices
- **AI, ML & Data Science:** Deep Learning (CNN/RNN), NLP, TensorFlow, Scikit-learn, OpenCV, Pandas, LangChain, LangGraph, Agentic AI, RAG Systems, Vector Databases
- **Core Competencies:** Data Structures, Object-Oriented Programming (OOP), DBMS, System Design, Operating Systems

ACHIEVEMENTS

- **Speed Quant Competition (First Place):** Secured **1st Rank** at NIT Jalandhar for solving complex algorithmic challenges under strict timed conditions, outperforming **200+ top participants**.
- **LeetCode:** Solved **750+ algorithmic challenges**, applying principles of Data Structures, dynamic programming, and edge-case testing, ranking in the **top 5%** of active users globally.
- **Codeforces (Specialist — 1502 Rating):** Solved **300+ logic problems** and maintained a highly competitive rating through consistent, high-speed performance in live global contest environments, including securing **Global Rank 676** in Codeforces Round 1104.

EXPERIENCE

- **Advance Application Engineering (AEH) Intern — AI Agent Development — Accenture, Bangalore — Jun 2026–Present:**
 - Architecting and deploying enterprise-grade **AI agents** leveraging **Retrieval-Augmented Generation (RAG)** pipelines to seamlessly ground Large Language Models (LLMs) in proprietary company data.
 - Generating and managing **vector embeddings** within scalable vector databases to enable high-speed semantic search, significantly improving context retrieval accuracy and reducing model hallucinations.
 - Integrating intelligent workflows via **Node.js** and Python backend services, while developing interactive **React.js** and **Tailwind CSS** interfaces for real-time monitoring of agent performance and user interactions.
- **CTP (NITJ Placement Portal) [Live Site] — Backend Lead — NIT Jalandhar — 2024–Present:**
 - Architected backend services using **Node.js** and **Express.js**, serving **1,000+ students** and **200+ recruiters** via **25+ RESTful APIs** for verification and automated interview scheduling workflows.
 - Optimized **MongoDB** indexing and implemented Role-Based Access Control (RBAC), accelerating data retrieval by **35%** and flawlessly processing over **10,000 daily API requests** during peak placement traffic.
 - Engineered responsive dashboards utilizing **React.js** and **Tailwind CSS**, improving user navigation speed by **20%**.
- **Xceed (NITJ Timetable) [Live Site] — Top Contributor — NIT Jalandhar — 2025–Present:**
 - Designed a scheduling engine utilizing **React.js** and **Tailwind CSS** with Google Maps API integration for real-time campus navigation, automating timetable generation for **5,000+ users** while eliminating **98% of scheduling conflicts**.
 - Reduced API latency by **40%** by optimizing **Express.js** backend data structures and **MongoDB** indexes across **10+ distinct departments**.
- **Third Space Travel (Travel Booking Platform) [Live Site] — Full-Stack Developer — On-Site — 2026:**
 - Developed a comprehensive travel booking platform for global clients using **React** and **TypeScript**, integrating **30+ RESTful endpoints** and utilizing **Prisma ORM** with **MySQL** to decrease average database query execution time by **50ms**.
 - Enabled online financial transactions by integrating the **PhonePe SDK** and **tRPC**, securely processing bookings with a **99.9% success rate** and reducing checkout drop-offs by **20%**.
 - Built frontend components using **SCSS** and Framer Motion to support cross-device rendering, increasing mobile user retention by **25%** and achieving a Lighthouse performance score of **95/100**.

PROJECTS

- **Shadow-Room: E2EE Secure Chat System [Live Demo] [Repo] — Sole Developer — 2026:**
 - Built an end-to-end encrypted (E2EE) messaging platform utilizing **Golang**, **React.js**, and **WebSockets** for low-latency, real-time asynchronous communication across distributed networks.
 - Implemented cryptographic handshakes (**RSA**, **MD5**, **SHA-256**) and the **Double Ratchet Algorithm** to establish forward secrecy, message integrity, and group scaling via Sender Keys.
 - Integrated multi-factor authentication and executed **PostgreSQL** and **Protocol Buffers** modifications to support high-throughput concurrent user sessions while ensuring zero data leaks.